EXP 5 : Simulation of LTE-Advanced Carrier Aggregation: Component Carriers and Throughput Comparison

MATLAB CODE:

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clc;
clear;
close all;

%% LTE-ADVANCED CARRIER AGGREGATION SIMULATION

% Component Carrier configuration
CC_BW = 20;                % Bandwidth per carrier in MHz
CC_values = [1 2 3 4 5];    % Number of component carriers

% System parameters
SINR_dB = 0:5:30;
overhead = 0.80;           % 80% effective bandwidth
maxEfficiency = 5;         % Maximum spectral efficiency (bps/Hz)

%% 1. COMPONENT CARRIER BANDWIDTH

totalBandwidth = CC_values * CC_BW;

figure('Name','Component Carrier Bandwidth');

bar(CC_values,totalBandwidth);

xlabel('Number of Component Carriers');
ylabel('Total Bandwidth (MHz)');

title('LTE-Advanced Component Carrier Aggregation');

grid on;

%% 2. THROUGHPUT CALCULATION

% Shannon capacity with practical efficiency limit
spectralEfficiency = min(log2(1 + 10.^(SINR_dB/10)), ...
                    maxEfficiency);

% Apply system overhead
spectralEfficiency = overhead * spectralEfficiency;
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% Throughput for different CC configurations
throughput = zeros(length(CC_values),length(SINR_dB));

for i = 1:length(CC_values)

    totalBW = CC_values(i)*CC_BW;

    throughput(i,:) = totalBW * ...
                      spectralEfficiency;

end

%% 3. THROUGHPUT VS SINR

figure('Name','Throughput vs SINR');

plot(SINR_dB,throughput(1,:),'LineWidth',2);
hold on;

plot(SINR_dB,throughput(2,:),'LineWidth',2);

plot(SINR_dB,throughput(3,:),'LineWidth',2);

plot(SINR_dB,throughput(4,:),'LineWidth',2);

plot(SINR_dB,throughput(5,:),'LineWidth',2);

xlabel('SINR (dB)');
ylabel('Throughput (Mbps)');

title('Throughput Comparison for Different CCs');

legend('1CC','2CC','3CC','4CC','5CC', ...
       'Location','northwest');

grid on;

%% 4. THROUGHPUT VS NUMBER OF CARRIERS

fixedSINR = 20;

fixedEfficiency = min(log2(1 + ...
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    10^(fixedSINR/10)),maxEfficiency);

fixedEfficiency = overhead*fixedEfficiency;

carrierThroughput = totalBandwidth * fixedEfficiency;

figure('Name','Carrier Aggregation Throughput');

bar(CC_values,carrierThroughput);

xlabel('Number of Component Carriers');
ylabel('Throughput (Mbps)');

title('Throughput Improvement Using Carrier Aggregation');

grid on;

%% 5. COMPONENT CARRIER RESOURCE ALLOCATION

figure('Name','Component Carrier Allocation');

hold on;

for i = 1:5

    rectangle('Position',[0,i,CC_BW,0.7], ...
        'FaceColor',[0.2 0.5 0.9]);

    text(CC_BW/2,i+0.35, ...
        ['CC ' num2str(i)], ...
        'HorizontalAlignment','center', ...
        'Color','white','FontWeight','bold');

end

xlabel('Frequency Bandwidth (MHz)');
ylabel('Component Carriers');

title('LTE-Advanced Component Carrier Allocation');

xlim([0 25]);
ylim([0 6]);
```

```
grid on;
hold off;

%% 6. THROUGHPUT COMPARISON AT DIFFERENT SINR

selectedSINR = [5 10 15 20 25 30];

selectedEfficiency = min(log2(1 + ...
   10.^(selectedSINR/10)),maxEfficiency);

selectedEfficiency = overhead*selectedEfficiency;

figure('Name','SINR Throughput Comparison');

bar(selectedSINR, ...
   (CC_values(5)*CC_BW)*selectedEfficiency);

xlabel('SINR (dB)');
ylabel('Throughput (Mbps)');

title('Maximum Carrier Aggregation Throughput vs SINR');

grid on;

%% 7. FINAL RESULTS AT 20 dB SINR

index = find(SINR_dB == 20);

fprintf('\n--- LTE-ADVANCED CARRIER AGGREGATION ---\n');

fprintf('SINR: %d dB\n',fixedSINR);

for i = 1:length(CC_values)

   fprintf('\nNumber of CCs: %d\n',CC_values(i));

   fprintf('Total Bandwidth: %.2f MHz\n', ...
      totalBandwidth(i));

   fprintf('Throughput: %.2f Mbps\n', ...
      throughput(i,index));

end
```

```
%% 8. THROUGHPUT IMPROVEMENT

baseThroughput = throughput(1,index);

fprintf('\n--- THROUGHPUT IMPROVEMENT ---\n');

for i = 1:length(CC_values)

    improvement = ...
        ((throughput(i,index)-baseThroughput) ...
        /baseThroughput)*100;

    fprintf('%d CC: %.2f %% improvement\n', ...
        CC_values(i),improvement);

end

%% 9. FINAL COMPARISON TABLE

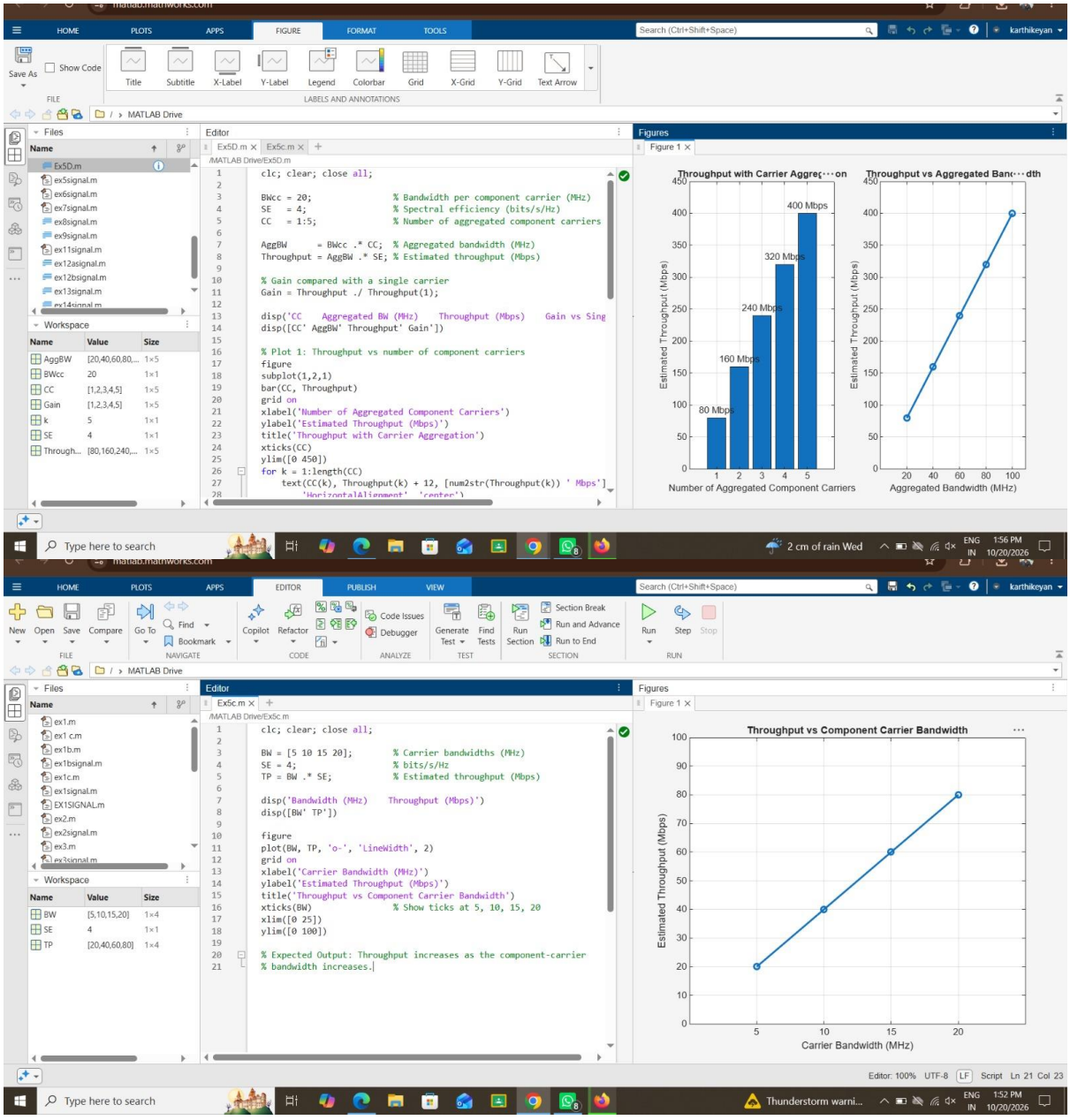
Results = table( ...
    CC_values', ...
    totalBandwidth', ...
    throughput(:,index), ...
    'VariableNames', ...
    {'Number_of_CC','Total_Bandwidth_MHz', ...
    'Throughput_Mbps'});

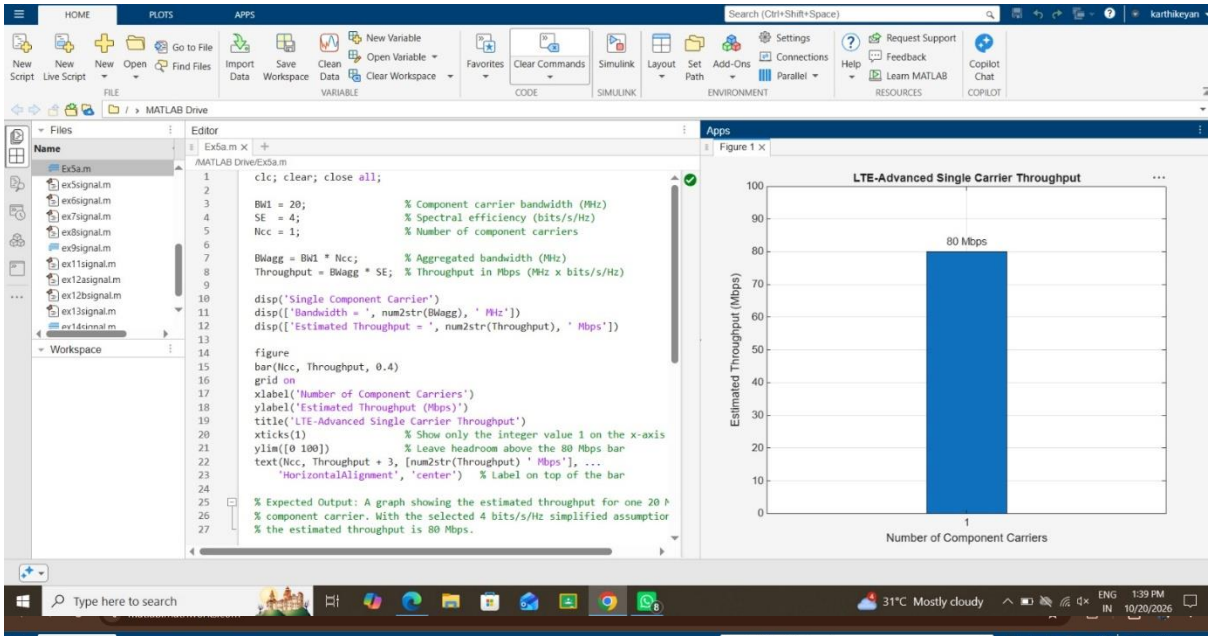
disp(' ');
disp('--- FINAL COMPARISON TABLE ---');

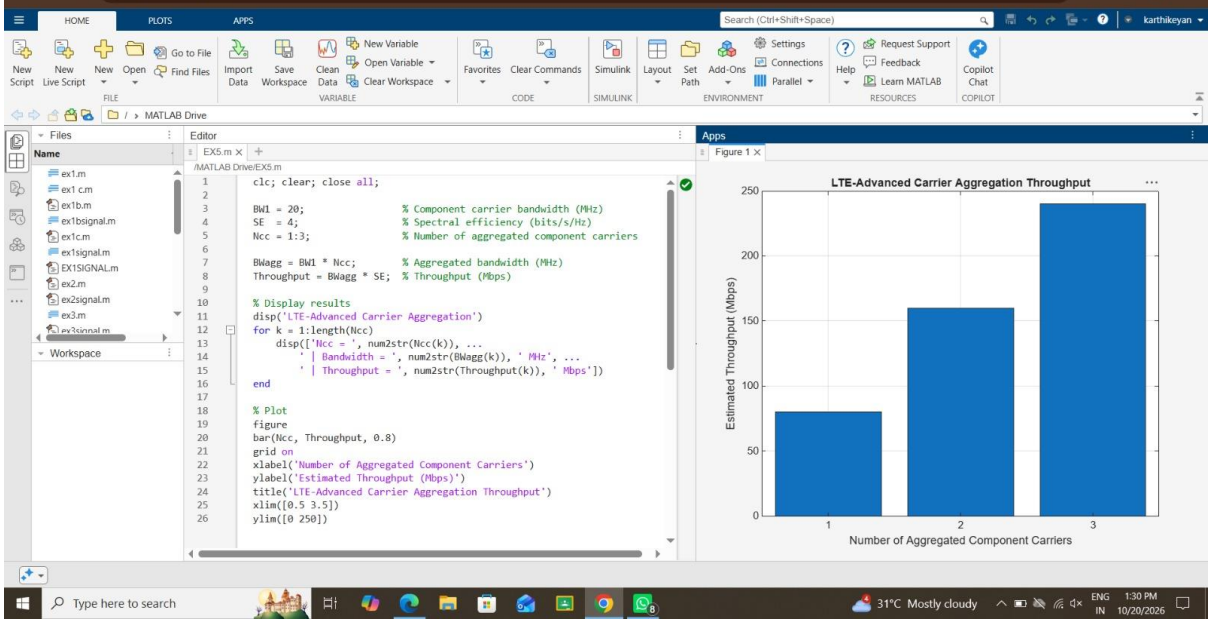
disp(Results);

fprintf('\nSimulation Completed Successfully!\n');
```

OUTPUT:


Throughput with Carrier Aggregation
Throughput vs Aggregated Bandwidth
Estimated Throughput (Mbps)
Number of Aggregated Component Carriers
Aggregated Bandwidth (MHz)
80 Mbps
160 Mbps
240 Mbps
320 Mbps
400 Mbps
Throughput vs Component Carrier Bandwidth
Carrier Bandwidth (MHz)


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ex9signal.m
ex11signal.m
ex12asignal.m
ex12bsignal.m
ex13signal.m
Workspace
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Ex5a.m
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clc; clear; close all;
BW1 = 20; % Component carrier bandwidth (MHz)
SE = 4; % Spectral efficiency (bits/s/Hz)
Ncc = 1; % Number of component carriers
BWagg = BW1 * Ncc; % Aggregated bandwidth (MHz)
Throughput = BWagg * SE; % Throughput in Mbps (MHz x bits/s/Hz)
disp('Single Component Carrier')
disp(['Bandwidth = ', num2str(BWagg), ' MHz'])
disp(['Estimated Throughput = ', num2str(Throughput), ' Mbps'])
figure
bar(Ncc, Throughput, 0.4)
grid on
xlabel('Number of Component Carriers')
ylabel('Estimated Throughput (Mbps)')
title('LTE-Advanced Single Carrier Throughput')
xticks(1) % Show only the integer value 1 on the x-axis
ylim([0 100]) % Leave headroom above the 80 Mbps bar
text(Ncc, Throughput + 3, [num2str(Throughput) ' Mbps'], ...
'HorizontalAlignment', 'center') % Label on top of the bar
% Expected Output: A graph showing the estimated throughput for one 20 M
% component carrier. With the selected 4 bits/s/Hz simplified assumption
% the estimated throughput is 80 Mbps.
Apps
Figure 1
LTE-Advanced Single Carrier Throughput
80 Mbps
Estimated Throughput (Mbps)
Number of Component Carriers
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ex1 c.m
ex1b.m
ex1bsignal.m
ex1c.m
ex1signal.m
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ex3.m
Workspace
Editor
EX5.m
/MATLAB Drive/EX5.m
clc; clear; close all;
BW1 = 20; % Component carrier bandwidth (MHz)
SE = 4; % Spectral efficiency (bits/s/Hz)
Ncc = 1:3; % Number of aggregated component carriers
BWagg = BW1 * Ncc; % Aggregated bandwidth (MHz)
Throughput = BWagg * SE; % Throughput (Mbps)
% Display results
disp('LTE-Advanced Carrier Aggregation')
for k = 1:length(Ncc)
disp(['Ncc = ', num2str(Ncc(k)), ...
' | Bandwidth = ', num2str(BWagg(k)), ' MHz', ...
' | Throughput = ', num2str(Throughput(k)), ' Mbps'])
end
% Plot
figure
bar(Ncc, Throughput, 0.8)
grid on
xlabel('Number of Aggregated Component Carriers')
ylabel('Estimated Throughput (Mbps)')
title('LTE-Advanced Carrier Aggregation Throughput')
xlim([0.5 3.5])
ylim([0 250])
Apps
Figure 1
LTE-Advanced Carrier Aggregation Throughput
Estimated Throughput (Mbps)
Number of Aggregated Component Carriers
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